Best Subset Selection

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Introduction

Best subset selection enumerates all 2^p possible predictor subsets and identifies the best subset by a chosen criterion (adjusted R^2, Mallows' C_p, BIC, cross-validation error). It is an exhaustive search and therefore avoids the sequence-dependence that plagues stepwise methods, but the combinatorial cost limits its practical use to modest p — typically up to about 30 predictors with branch-and-bound algorithms (`leaps`).

Prerequisites

A working understanding of multiple regression, model-selection criteria, and the bias-variance trade-off in choosing model complexity.

Theory

For each subset size $k \in \{0, 1, \ldots, p\}$, find the subset with the lowest residual sum of squares (equivalently, highest R^2). Then compare across sizes using a criterion that penalises complexity:

- **Adjusted R^2**: $1 - (1 - R^2)(n - 1)/(n - k - 1)$; rewards fit per parameter.
- **Mallows' C_p**: $\mathrm{RSS}_k/\hat{\sigma}^2 - n + 2k$; near k when the subset is adequate.
- **AIC**: $-2\ell + 2k$; favours predictive accuracy.
- **BIC**: $-2\ell + k \log n$; more parsimonious than AIC.
- **Cross-validation**: directly estimates predictive error.

`leaps::regsubsets` uses branch-and-bound to avoid evaluating every subset explicitly, making moderate-p best-subset feasible.

Assumptions

Standard linear-regression assumptions; criteria like AIC, BIC assume Gaussian residuals (other forms exist for GLMs).

R Implementation

```r
library(leaps)

d <- mtcars
fit <- regsubsets(mpg ~ ., data = d, nvmax = 5)
sum_fit <- summary(fit)

data.frame(
  n_pred = 1:5,
  adj_R2 = sum_fit$adjr2,
  Cp     = sum_fit$cp,
  BIC    = sum_fit$bic
)
```

```r
# Show the best subset at each size
sum_fit$which

# Plot the criterion
plot(fit, scale = "bic")
```

Output & Results

`regsubsets()` returns the best subset of each size up to `nvmax`. The summary provides criterion values; `sum_fit$which` shows which predictors are included at each size; the plot displays subsets ordered by the chosen criterion.

Interpretation

A reporting sentence: "Best-subset selection on `mtcars` (with `nvmax = 5`) identified weight (`wt`), quarter-mile time (`qsec`), and transmission (`am`) as the preferred three-predictor model under BIC (BIC $= -84$); cross-validated mean squared error confirmed this as the best subset, with substantially worse performance for both smaller and larger models." Always state the criterion and pair with cross-validation when stakes are high.

Practical Tips

- Best subset is computationally feasible up to about $p = 30$ with branch-and-bound; beyond that, switch to penalised methods (lasso, elastic net).
- Lasso and elastic net often perform similarly to best-subset selection with better scalability and less over-fitting; benchmark all three for high-dimensional problems.
- Like stepwise selection, best-subset chooses by criteria computed on the same data — cross-validate to assess true predictive performance.
- `regsubsets()` returns only the *single* best subset per size; near-ties can be missed and selection becomes unstable across resamples. Bootstrap to assess stability.
- Pre-specification of a subset based on substantive theory generally beats any data-driven search; reserve best-subset for genuinely exploratory work.
- For GLMs, use `bestglm::bestglm` or stepwise on AIC; pure best-subset enumeration is OLS-specific in `leaps`.

R Packages Used

`leaps::regsubsets` for branch-and-bound best-subset on linear models; `bestglm::bestglm` for GLMs; `glmnet` for lasso/elastic-net alternatives that scale to high p; `caret` and `tidymodels` for cross-validated subset comparison; `MuMIn::dredge` for model-averaging extensions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring

- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI